P48 - IS THIS WATER SAFE TO DRINK?

Water Potability Classification using a Support Vector Machine (SVM)

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Abstract

This project explores supervised machine learning to classify water samples as potable (1) or not potable (0) from nine chemical and physical measurements: pH, hardness, solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes and turbidity. The workflow includes data cleaning with median imputation, exploratory data analysis, feature engineering, feature selection, scaling and Support Vector Machine (SVM) comparison and evaluation.

Problem Statement

The objective is to predict the Potability label from the recorded measurements and evaluate the classifier. **Precision for 'safe'** receives special attention because a false positive means unsafe water is called safe, which would be dangerous in practice. This is an educational project, not a water-safety tool, and its output must never be used to decide whether water is safe to drink.

Objectives

- Inspect and clean the supplied CSV dataset, including missing values.
- Explore class balance, missing data, outliers and correlations.
- Engineer a new feature and select the most informative inputs.
- Split data before fitting the scaler to reduce leakage risk.
- Compare SVM kernels (linear, RBF) and regularisation strengths (C).
- Evaluate the selected model using multiple metrics and discuss why scores are modest.

This report was generated from water_potability.csv using the D55_P48 notebook. Results use a stratified 80/20 train-test split with random_state=42.

1. Dataset Description and Data Dictionary

The CSV contains **3,276 rows and 10 columns**: nine numeric water-quality measurements and the Potability target. The target contains 1,998 not-potable (0) and 1,278 potable (1) records. Three columns had missing values: ph (491), Sulfate (781) and Trihalomethanes (162). These were filled with column medians. One engineered feature, Mineral_Content, was added, giving 3,276 rows and 11 columns.

Class distribution: not potable = 61.0%; potable = 39.0%. **Duplicate rows:** 0. **Missing values after imputation:** 0.

Field descriptions

Field	Meaning	Type	Decision
ph	Measures water acidity or alkalinity.	Number	Keep (491 missing, median-filled)
Hardness	Measures water hardness.	Number	Keep
Solids	Total dissolved solids in water.	Number	Keep
Chloramines	Amount of chloramines in water.	Number	Keep
Sulfate	Sulfate concentration in water.	Number	Keep (781 missing, median-filled)
Conductivity	Ability of water to conduct electricity.	Number	Keep
Organic_carbon	Organic carbon concentration in water.	Number	Keep
Trihalomethanes	Trihalomethanes concentration in water.	Number	Keep (162 missing, median-filled)
Turbidity	Measures cloudiness of water.	Number	Keep
Potability	Target: 0 = not potable, 1 = potable.	Number (0/1)	Keep (target)
Mineral_Content	Engineered feature: Hardness + Sulfate.	Number	Keep (engineered)

Why scores are modest: the features have very weak individual relationships with Potability (see Section 2), so even a flexible model has little signal to learn from. Low scores in this project are an expected property of the data, not only of the model.

The feature names follow the naming pattern in the supplied dataset; field descriptions are based on those names.

2. Exploratory Data Analysis (EDA)

EDA was used to inspect the potability class balance, the amount of missing data, the correlations between measurements and the target, and the strength of each feature.

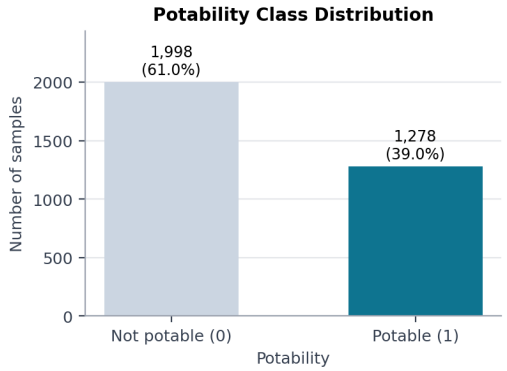


Figure 1. Potability class counts.

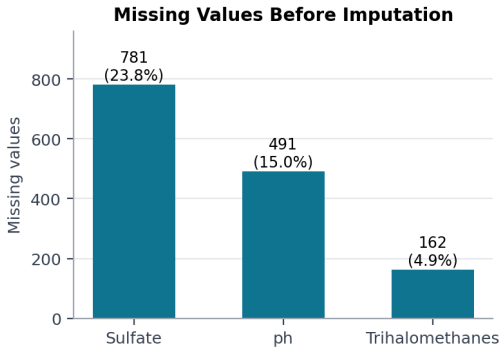


Figure 2. Missing values before median imputation.

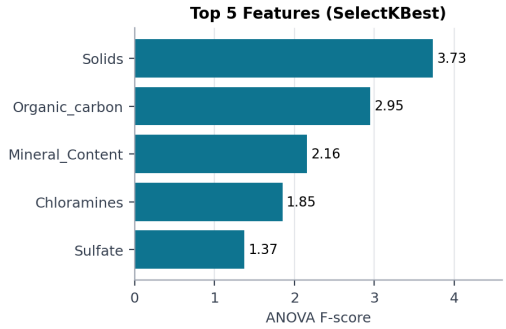


Figure 3. ANOVA F-scores of the five selected features.

> **Takeaway:** even the best feature (Solids) has an F-score of only 3.7, and the correlations in Figure 4 are all close to zero. No single measurement separates safe from unsafe water well.

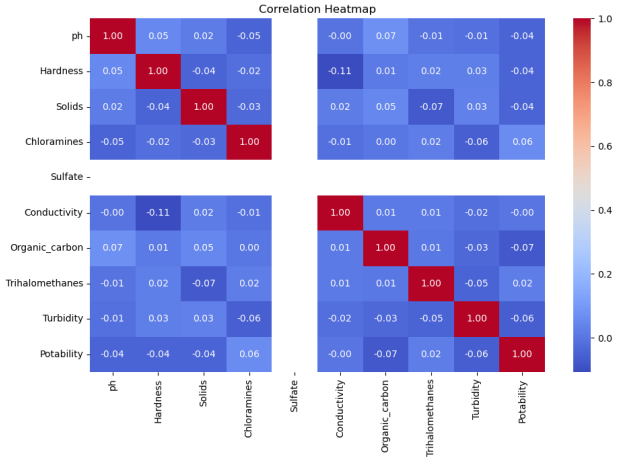


Figure 4. Correlation heatmap from the outlier-filtered EDA copy (629 rows). The blank Sulfate row means no variation remained after filtering.

Five EDA insights

1. Not-potable samples outnumber potable ones: 1,998 vs 1,278 (61.0% vs 39.0%), a mild class imbalance.
2. Sulfate has the most missing values (781, 23.8%), followed by ph (491, 15.0%) and Trihalomethanes (162, 4.9%); all were filled with medians.
3. Linear relationships with Potability are very weak: in the filtered EDA copy the largest absolute correlation is only 0.07 (Organic_carbon), with Chloramines at 0.06 and Turbidity at -0.06.
4. Iterative IQR outlier removal discarded 2,647 of 3,276 rows (80.8%), leaving 629. This was far too aggressive, so it was used only for EDA; modelling used the full cleaned table.
5. Feature scores are low: the best ANOVA F-score is 3.73 (Solids), followed by Organic_carbon (2.95) and Mineral_Content (2.16).

3. Methodology and Model Comparison

Cleaning: no duplicate rows were found; missing ph, Sulfate and Trihalomethanes values were filled with column medians. Added Mineral_Content = Hardness + Sulfate.

Train-test split: stratified 80/20 split: 2,620 training records (1,598 not potable, 1,022 potable) and 656 testing records (400 / 256), random_state=42. StandardScaler was fitted on the training data only.

Feature selection and modelling: SelectKBest with an ANOVA F-test kept five features: Solids, Chloramines, Sulfate, Organic_carbon and Mineral_Content. An SVC with default settings (RBF kernel, C=1) was the baseline, then a linear kernel and RBF with C = 0.1, 1 and 10 were compared.

Held-out test-set comparison

Setting	Accuracy	Precision	Recall	F1	Specificity	AUC
rbf, C=0.1 (selected)	0.619	0.750	0.035	0.067	0.992	0.586
rbf, C=1 (baseline)	0.637	0.705	0.121	0.207	0.968	0.608
rbf, C=10	0.642	0.636	0.191	0.294	0.930	0.606
linear, C=1	0.610	0.000	0.000	0.000	1.000	0.511

The notebook lists the baseline three times (Baseline SVM, kernel=rbf, rbf C=1) because they are the same model; it is shown once. The linear kernel predicted no potable samples, so its precision is reported as 0.

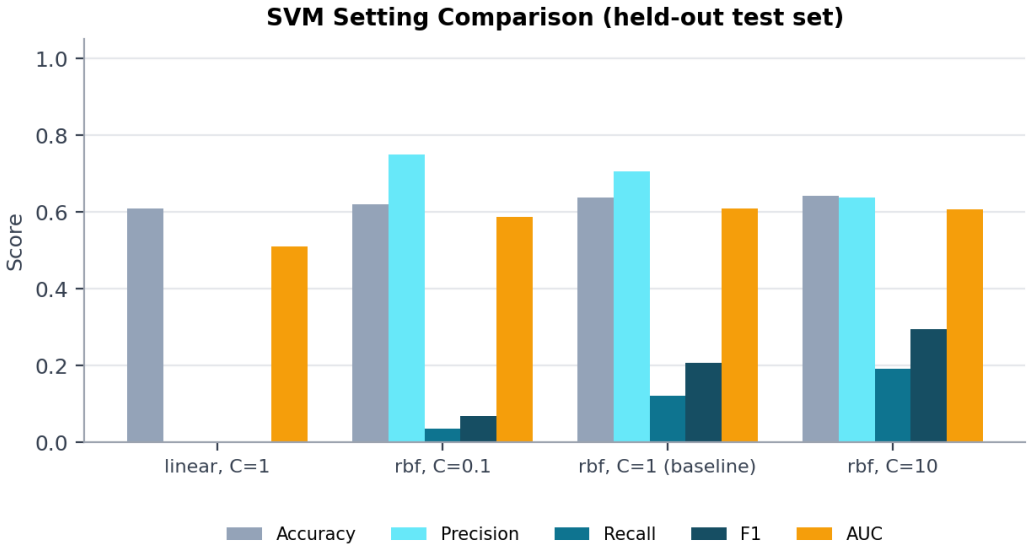


Figure 5. Metric comparison across SVM settings.

> **Selection rule:** highest precision for 'potable'. Selected setting: **SVC, kernel=rbf, C=0.1**. This setting has the best precision (0.750), but it labels almost every sample unsafe (recall 0.035), so the high precision rests on very few positive predictions (see Section 4).

4. Final Model Evaluation

The selected model (SVC, rbf kernel, C=0.1) was evaluated on 656 held-out test records. It classified 406 records correctly and made 250 errors, almost all of them potable water labelled as not potable.

61.89%	75.00%	3.52%	0.067	0.586
Accuracy	Precision (potable)	Recall (potable)	F1-score	ROC-AUC

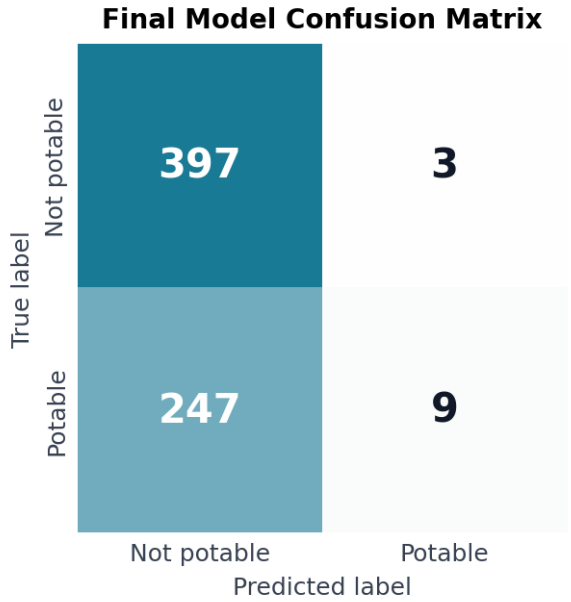


Figure 6. Confusion matrix (counts reconstructed from the reported metrics).

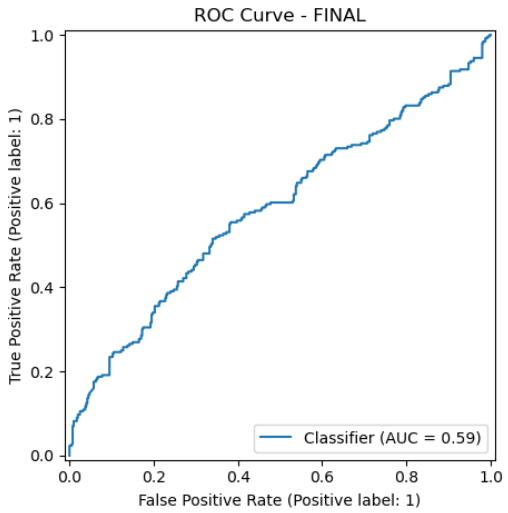


Figure 7. ROC curve (notebook output).

Metric	Result	Meaning
Accuracy	61.89%	Share of all test records classified correctly.
Precision (potable)	75.00%	Share of samples predicted potable that really were potable.
Recall (potable)	3.52%	Share of truly potable samples identified.
F1-score	0.0672	Harmonic mean of precision and recall.
Specificity	99.25%	Share of truly not-potable samples correctly identified.
ROC-AUC	0.5857	Ranking separation across thresholds (0.5 = no better than chance).

Confusion matrix counts (derived): TN = 397, FP = 3, FN = 247, TP = 9. Only 12 samples were predicted potable; 3 of them were actually unsafe.

> **Read with care:** always predicting 'not potable' already gives 61.0% accuracy, so 61.9% is barely better. The 75% precision comes from 9 correct out of 12 predictions, which is too few to be reliable, and 1 in 4 'safe' calls was wrong. Counts are reconstructed from the notebook's reported metrics and match its accuracy exactly.

5. Illustrative Predictions, Conclusion and Limitations

The three examples below are the first three rows of the dataset, passed unchanged through the scaler and final model, following the notebook's procedure. They are teaching examples, not real safety checks.

Example	Solids	Chloramines	Sulfate	Organic C	Mineral	Predicted label
Sample 1	20,791	7.30	368.5	10.38	573.4	0 - Not potable
Sample 2	18,630	6.64	333.1	15.18	462.5	0 - Not potable
Sample 3	19,910	9.28	333.1	16.87	557.3	0 - Not potable

Sulfate 333.1 in samples 2 and 3 is the imputed median. Mineral = Mineral_Content. The notebook prints labels only, so no probabilities are reported, and the true labels of these rows were not shown.

Conclusion

The project demonstrates a supervised ML workflow from cleaning and EDA through feature engineering, scaling and SVM tuning. The selected setting (rbf kernel, C=0.1) reached 61.89% accuracy, 75.00% precision for potable water and 0.586 AUC on the held-out test set. The weak feature-target relationships limit every setting, and the selected model is conservative: it calls almost everything unsafe. These results apply to this split only and do not show a usable water-safety model.

Limitations

- This is an educational project, not a water-safety system. Never use it to judge real drinking water.
- Selection by precision alone favoured a model with recall of 3.5%; precision is based on only 12 positive predictions.
- Accuracy (61.9%) is close to the always-'not potable' baseline (61.0%), and AUC (0.586) is near chance.
- Median imputation and feature selection were run on the full dataset before the train-test split, which is a mild leakage risk.
- Iterative IQR outlier removal (80.8% of rows) was too aggressive and was used for EDA only.
- One train-test split was used; cross-validation, class weighting and a precision-recall threshold study were not performed.
- Illustrative examples are existing dataset rows, not new real-world samples.

Submission reminder

Check the report against the instructor's rubric, and ensure the final notebook is run from top to bottom with the same CSV file before submission.